

Q1. A company conducted a survey to determine the preferred mode of transportation among employees as shown below:

transport = ["Train", "Bicycle", "Bike", "Car", "Bus"],

employees = [220, 60, 70, 80, 40]

Create a Pie Chart to represent the percentage of employees using each mode of transportation.

Display the percentage contribution of each category. Give appropriate labels and colors.

[10 Marks]

Q2. Using the given housing dataset, develop a Random Forest Regression model to predict house prices based on the available property attributes. Perform the necessary data preprocessing, train and test the model, and evaluate its predictive performance using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

OR

[20 Marks]

Q2. Create the following dataset in python. Apply the Apriori algorithm to discover frequently used learning resources and generate association rules among them. Repeat the process with different minimum support values 0.3, and 0.4.

[20 Marks]

TID	Items
1	Video Lecture, Quiz
2	Video Lecture, Assignment
3	Quiz, Discussion Forum
4	Video Lecture, Quiz, Assignment
5	Assignment, Discussion Forum
6	Video Lecture, Certificate
7	Quiz, Certificate
8	Video Lecture, Quiz
9	Assignment, Certificate
10	Video Lecture, Quiz, Certificate

Q3. Viva

[5 Marks]